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Mathematics for Data Science

MATH101 | Suggested sequence · Semester 1 | Student-authored study guide

How to use these notes

Linear algebra, functions, derivatives, gradients, and optimization concepts used by data methods. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Vectors and matrices

A vector is an ordered list of values; a matrix is a rectangular array. Dot products combine aligned dimensions; matrix multiplication requires the inner dimensions to match. Shapes are a practical correctness check.

Worked example

```
[1,2]·[3,4] = 1×3 + 2×4 = 11.
```

Check your understanding

- When is A×B defined?
- What does a dot product measure geometrically?

MODULE 02 · Functions and derivatives

A function maps inputs to outputs. A derivative measures local rate of change; partial derivatives vary one input while holding others fixed. The chain rule differentiates compositions.

Worked example

```
For f(x)=x², f\u2032(x)=2x, so the slope at x=3 is 6.
```

Check your understanding

- What does a derivative represent?
- When is the chain rule useful?

MODULE 03 · Gradients and optimization

The gradient collects partial derivatives. Gradient descent steps opposite the gradient to reduce a differentiable objective. Learning rate controls step size; poor scaling or excessive step size can prevent convergence.

Worked example

```
For squared error, iteratively update parameters using ;,    ¸ "  ;r t¢ƒ¸).
```

Check your understanding

- Why step opposite the gradient?
- What happens when the learning rate is too large?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.